

```

%root_path = fileparts(mfilename("fullpath"));
%root_path = pwd;

% retrieve selected cuts used for further fitting
[GH_FFCutsFits,CutFitFileGH] =
recover_selected_cuts('GH_FFCutsFits','e:\SHARE\Fe\Data\analysis\07_fit_with_multicu
t_all\sel_cuts');

```

```

fn = fieldnames(GH_FFCutsFits);
%nf = 11
%for nf = 2:numel(fn)
%selectCuts = fn{nf};
selectCuts = 'Ei400_off200GHmidFF';
if ismember(selectCuts(1:3),{'msk','dat'})||...
    contains(selectCuts,'all_fit_par')
    %continue
    error('Can not fit field: %s',selectCuts);
end

fprintf('*****\n');

```

```

*****

```

```

fprintf('Processing cuts: %s \n', selectCuts);

```

```

Processing cuts: Ei400_off200GHmidFF

```

```

fprintf('*****\n');

```

```

*****

```

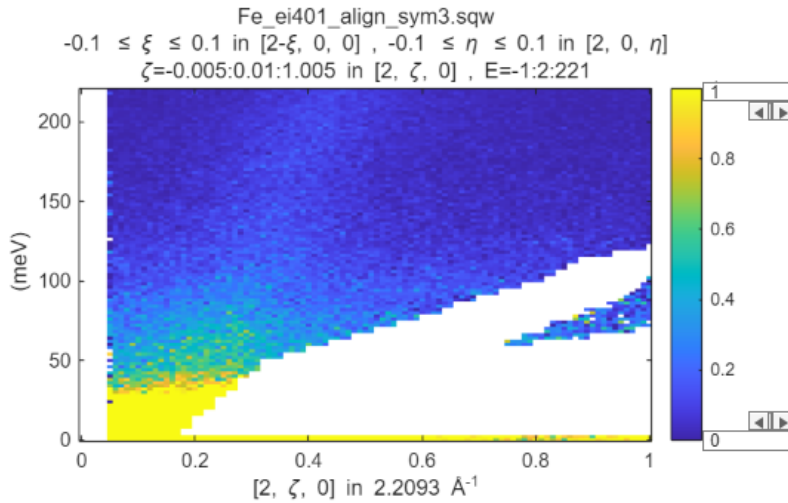
```

cuts_list = GH_FFCutsFits.(selectCuts);
if ~iscell(cuts_list)
    cuts_list = {cuts_list};
end
if contains(selectCuts,'Ei200')
    min_Ei = 30;
    mi = maps_instrument(200,600,'S');
end
if contains(selectCuts,'Ei400')
    min_Ei = 40;
    mi = maps_instrument(401,600,'S');
end
if contains(selectCuts,'Ei800')
    min_Ei = 70;
    mi = maps_instrument(786,600,'S');
end
sample=IX_sample(true,[1,0,0],[0,1,0],'cuboid',[0.04,0.03,0.02]);
for j=1:numel(cuts_list)

```

```
cuts_list{j} = cuts_list{j}.set_instrument(mi);
cuts_list{j} = cuts_list{j}.set_sample(sample);
plot(cuts_list{j}); lz 0 1; keep_figure;
```

end



```
dir_name = "GH";

dE_step = 2; %original energy transfer step data were binned to. No point in going
finer
half_dE = 5; % half width of data binning

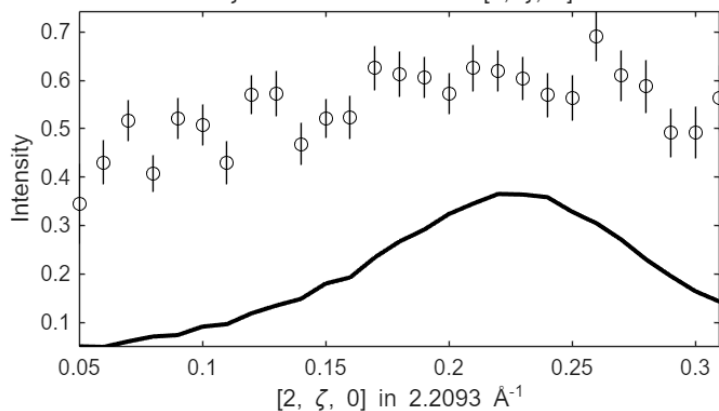
en_mm = cuts_list{1}.data.img_range(:,4);
en_mm(1) = max(min_Ei,en_mm(1));
%en_mm(1) = 100;
%en_mm(2) = 170;
cut_en = en_mm(1)+half_dE:2*half_dE:(en_mm(2)-half_dE);

cut_dimensions = 1;
GH_FFcutsFits.(selectCuts) = cuts_list;

GH_FFcutsFits = fit_multicutsFF_along_direction(...
    GH_FFcutsFits,selectCuts,cut_dimensions,cut_en,dE_step,half_dE,true);
```

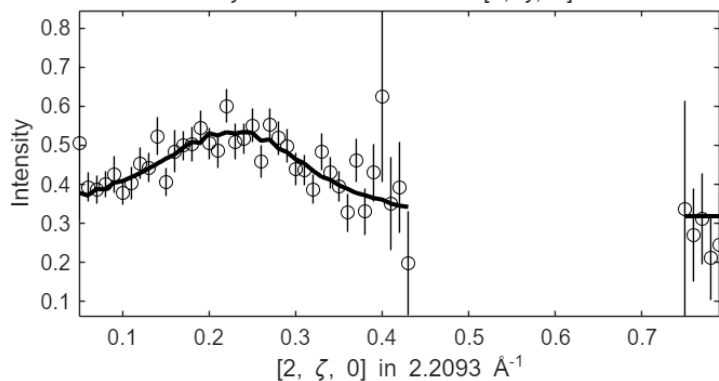
```
*****
**** En = 45±5 going to 215
*****
```

Fe_ei401_align_sym3.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, \eta]$, $40 \leq E \leq 100$ eV
 $\zeta = -0.005:0.01:1.005$ in $[2, \zeta, 0]$

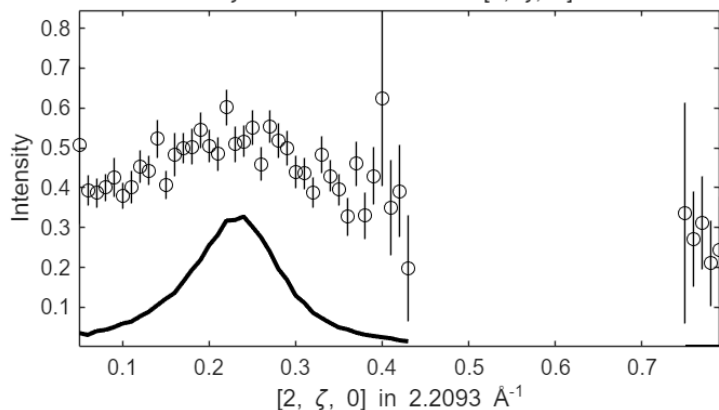


**** En = 55±5 going to 215

Fe_ei401_align_sym3.sqw
 $S = 1.4 \pm 0.8$; $J_0 = 27 \pm 3.6$; $J_1 = 0 \pm 0$; $\gamma = 29.1302 \pm 22.3047$
 $-0.1 \leq \xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, \eta]$, $50 \leq E \leq 100$ eV
 $\zeta = -0.005:0.01:1.005$ in $[2, \zeta, 0]$



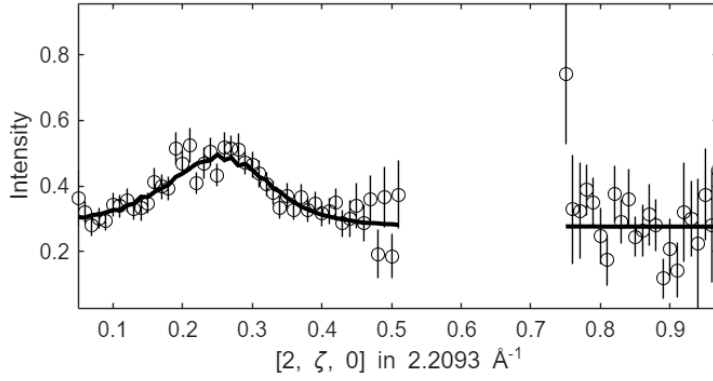
Fe_ei401_align_sym3.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, \eta]$, $50 \leq E \leq 100$ eV
 $\zeta = -0.005:0.01:1.005$ in $[2, \zeta, 0]$



**** En = 65±5 going to 215

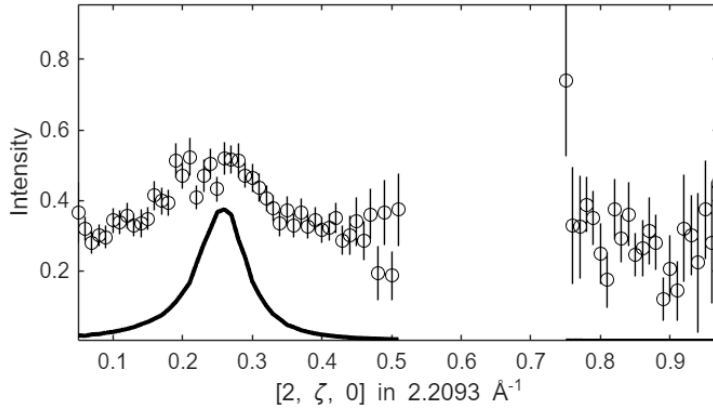
Fe_ei401_align_sym3.sqw

S= 1 ± 0.061 ; J0= 25 ± 0.8 ; J1= 0 ± 0 ; gamma= 17.7904 ± 0.878408
 $-0.1 \leq \xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, \eta]$, $60 \leq E \leq 70$ eV
 $\zeta = -0.005:0.01:1.005$ in $[2, \zeta, 0]$



Fe_ei401_align_sym3.sqw

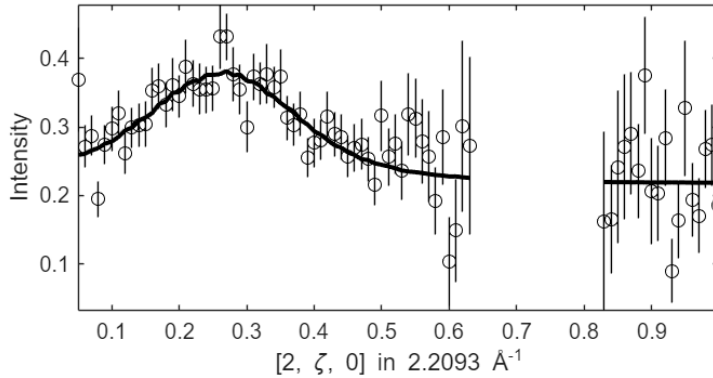
$-0.1 \leq \xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, \eta]$, $60 \leq E \leq 70$ eV
 $\zeta = -0.005:0.01:1.005$ in $[2, \zeta, 0]$



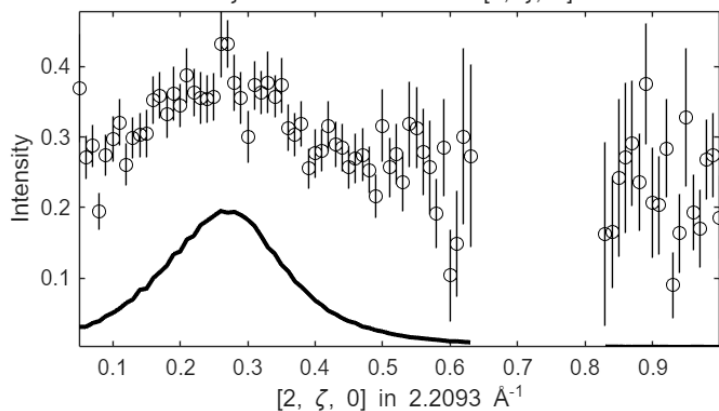
**** En = 75 ± 5 going to 215

Fe_ei401_align_sym3.sqw

S= 3 ± 3 ; J0= 42 ± 26 ; J1= 0 ± 0 ; gamma= 142.196 ± 209.529
 $-0.1 \leq \xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, \eta]$, $70 \leq E \leq 80$ eV
 $\zeta = -0.005:0.01:1.005$ in $[2, \zeta, 0]$

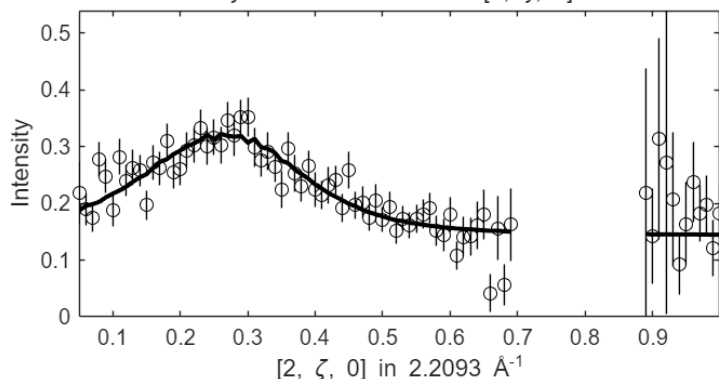


Fe_ei401_align_sym3.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, \eta]$, $70 \leq E \leq 100$ eV
 $\zeta = -0.005:0.01:1.005$ in $[2, \zeta, 0]$

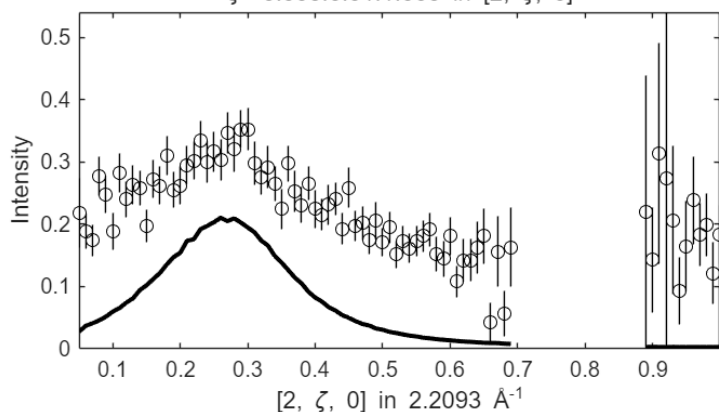


**** En = 85±5 going to 215

Fe_ei401_align_sym3.sqw
 $S = 6.2 \pm 11$; $J_0 = 68 \pm 96$; $J_1 = 0 \pm 0$; $\gamma = 369.147 \pm 1106.83$
 $-0.1 \leq \xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, \eta]$, $80 \leq E \leq 100$ eV
 $\zeta = -0.005:0.01:1.005$ in $[2, \zeta, 0]$



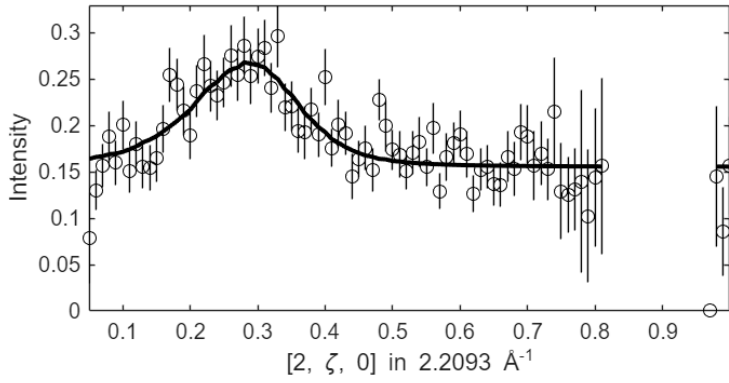
Fe_ei401_align_sym3.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, \eta]$, $80 \leq E \leq 100$ eV
 $\zeta = -0.005:0.01:1.005$ in $[2, \zeta, 0]$



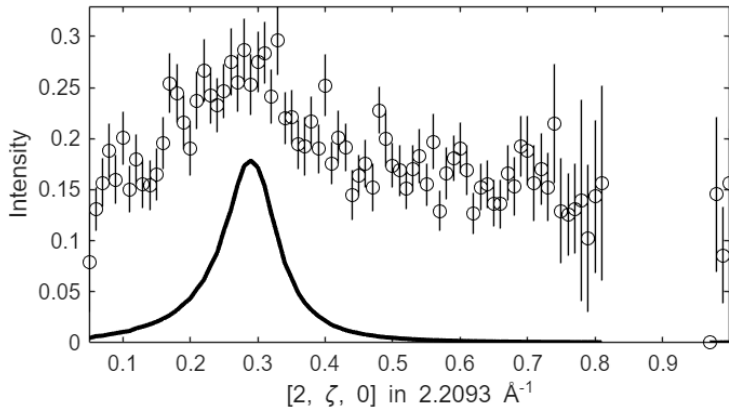
**** En = 95±5 going to 215

Warning: Dataset:1 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 0.9901

Fe_ei401_align_sym3.sqw
 $S=0.75\pm0.16$; $J_0= 30\pm 1.5$; $J_1= 0\pm 0$; $\gamma=27.9373\pm11.7159$
 $-0.1 \leq \xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, \eta]$, $90 \leq E \leq 1$
 $\zeta=-0.005:0.01:1.005$ in $[2, \zeta, 0]$



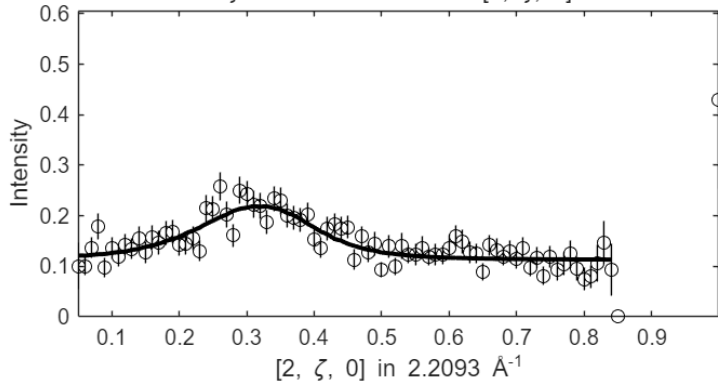
Fe_ei401_align_sym3.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, \eta]$, $90 \leq E \leq 1$
 $\zeta=-0.005:0.01:1.005$ in $[2, \zeta, 0]$



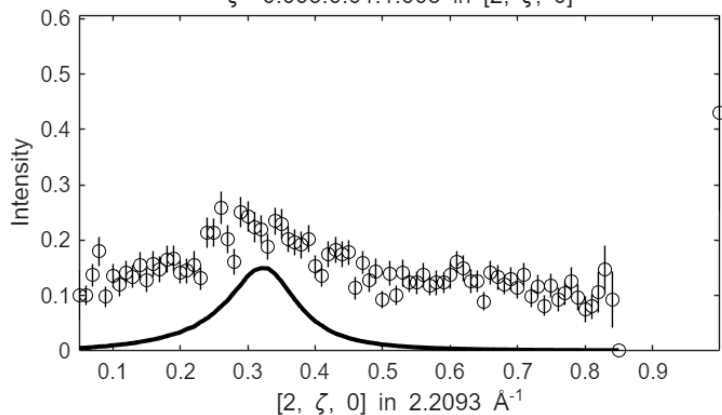
 **** En = 105±5 going to 215

Warning: Dataset:1 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 0.9901

Fe_ei401_align_sym3.sqw
 $S=0.86\pm0.052$; $J_0= 28\pm 1$; $J_1= 0\pm 0$; $\gamma=39.1067\pm2.06223$
 $0.1 \leq \xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, \eta]$, $100 \leq E \leq$
 $\zeta=-0.005:0.01:1.005$ in $[2, \zeta, 0]$

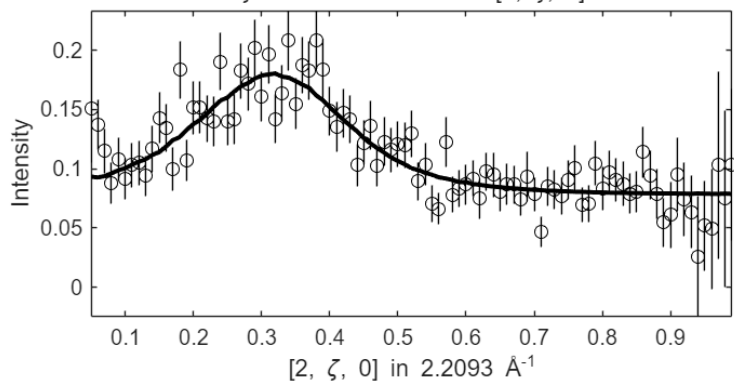


Fe_ei401_align_sym3.sqw
 $0.1 \leq \xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, \eta]$, $100 \leq E \leq$
 $\zeta=-0.005:0.01:1.005$ in $[2, \zeta, 0]$

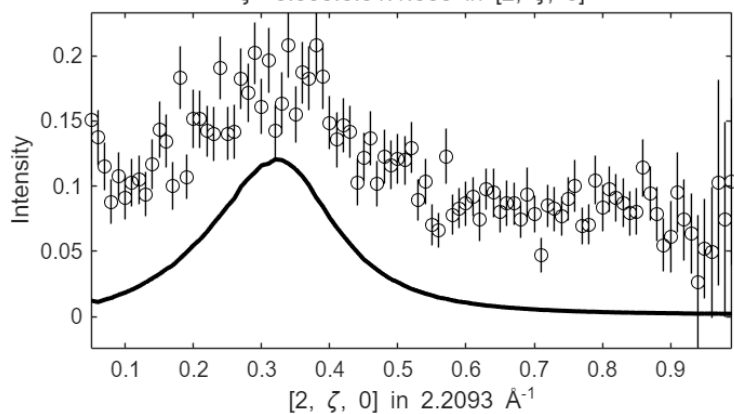


**** En = 115±5 going to 215

Fe_ei401_align_sym3.sqw
 $S = 1.7 \pm 0.6$; $J_0 = 37 \pm 6$; $J_1 = 0 \pm 0$; $\gamma = 111.471 \pm 54.4956$
 $0.1 \leq \xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, \eta]$, $110 \leq E \leq$
 $\zeta=-0.005:0.01:1.005$ in $[2, \zeta, 0]$

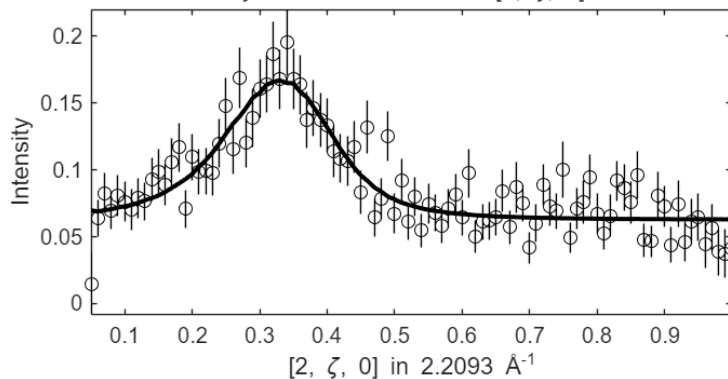


Fe_ei401_align_sym3.sqw
 $0.1 \leq \xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, \eta]$, $110 \leq E \leq$
 $\zeta=-0.005:0.01:1.005$ in $[2, \zeta, 0]$

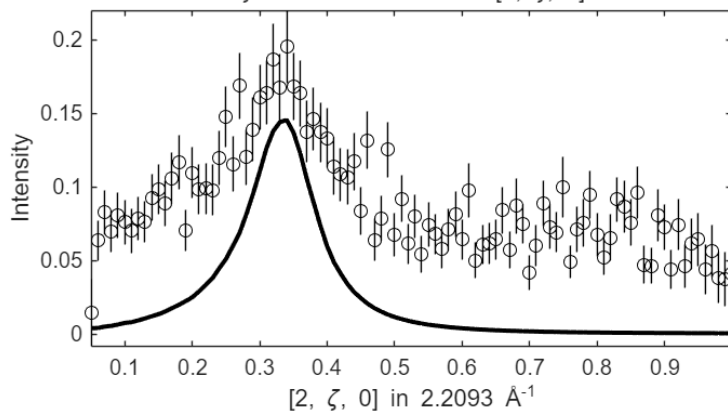


**** En = 125±5 going to 215

Fe_ei401_align_sym3.sqw
 $S=0.88\pm0.043$; $J_0= 31\pm0.85$; $J_1= 0\pm 0$; $\gamma=40.1573\pm1.79175$
 $0.1 \leq \xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, \eta]$, $120 \leq E \leq 120$
 $\zeta=-0.005:0.01:1.005$ in $[2, \zeta, 0]$

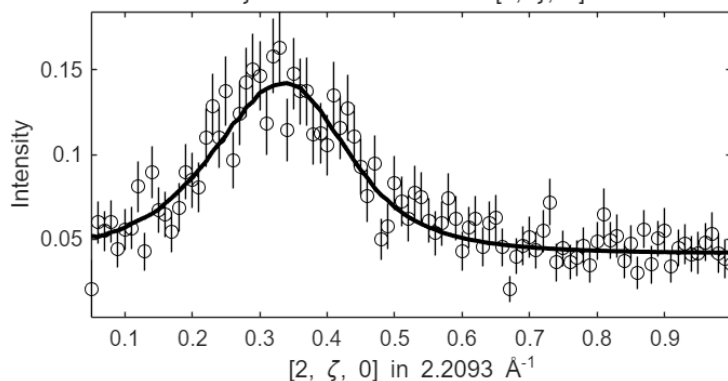


Fe_ei401_align_sym3.sqw
 $0.1 \leq \xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, \eta]$, $120 \leq E \leq 120$
 $\zeta=-0.005:0.01:1.005$ in $[2, \zeta, 0]$

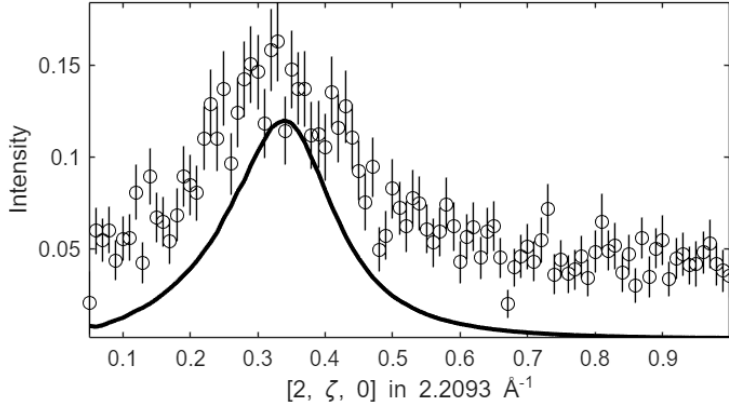


 **** En = 135±5 going to 215

Fe_ei401_align_sym3.sqw
 $S= 1.5\pm0.24$; $J_0= 36\pm 2.3$; $J_1= 0\pm 0$; $\gamma=86.5656\pm21.3611$
 $0.1 \leq \xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, \eta]$, $130 \leq E \leq 130$
 $\zeta=-0.005:0.01:1.005$ in $[2, \zeta, 0]$

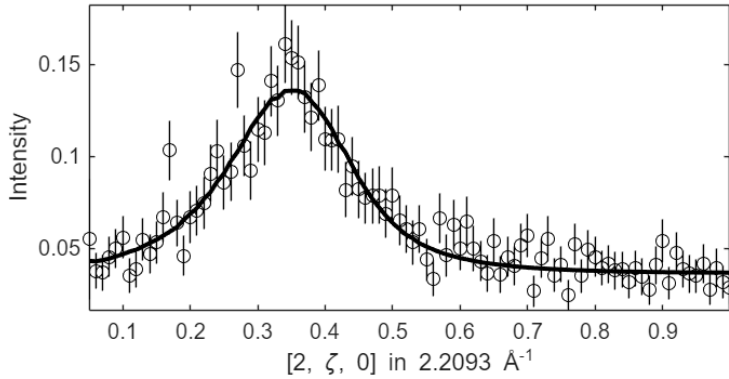


Fe_ei401_align_sym3.sqw
 $0.1 \leq \xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, \eta]$, $130 \leq E \leq 140$ eV
 $\zeta = -0.005:0.01:1.005$ in $[2, \zeta, 0]$

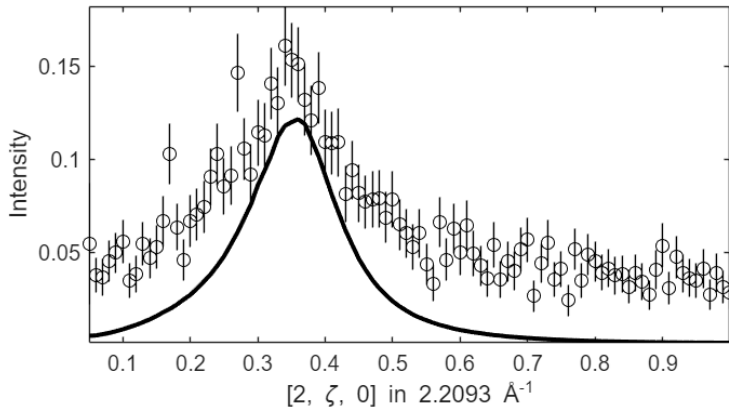


 **** En = 145±5 going to 215

Fe_ei401_align_sym3.sqw
 $S = 1.2 \pm 0.044$; $J_0 = 34 \pm 0.62$; $J_1 = 0 \pm 0$; $\gamma = 66.5415 \pm 2.02127$
 $0.1 \leq \xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, \eta]$, $140 \leq E \leq 150$ eV
 $\zeta = -0.005:0.01:1.005$ in $[2, \zeta, 0]$

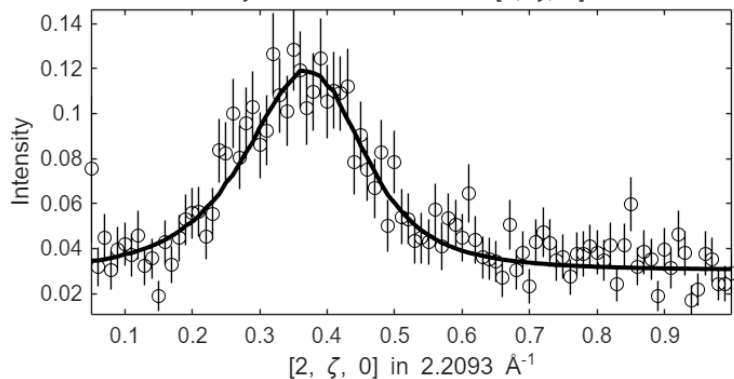


Fe_ei401_align_sym3.sqw
 $0.1 \leq \xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, \eta]$, $140 \leq E \leq 150$ eV
 $\zeta = -0.005:0.01:1.005$ in $[2, \zeta, 0]$

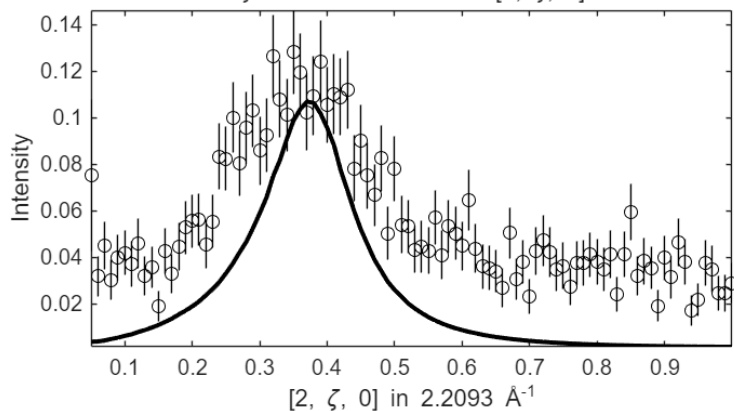


 **** En = 155±5 going to 215

Fe_ei401_align_sym3.sqw
 $S = 1 \pm 0.11$; $J_0 = 33 \pm 1$; $J_1 = 0 \pm 0$; $\gamma = 62.9826 \pm 10.9409$
 $0.1 \leq \xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, \eta]$, $150 \leq E \leq 150$
 $\zeta = -0.005:0.01:1.005$ in $[2, \zeta, 0]$

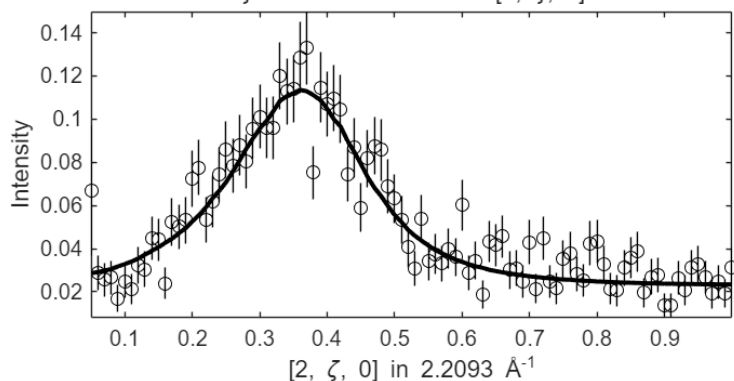


Fe_ei401_align_sym3.sqw
 $0.1 \leq \xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, \eta]$, $150 \leq E \leq 150$
 $\zeta = -0.005:0.01:1.005$ in $[2, \zeta, 0]$

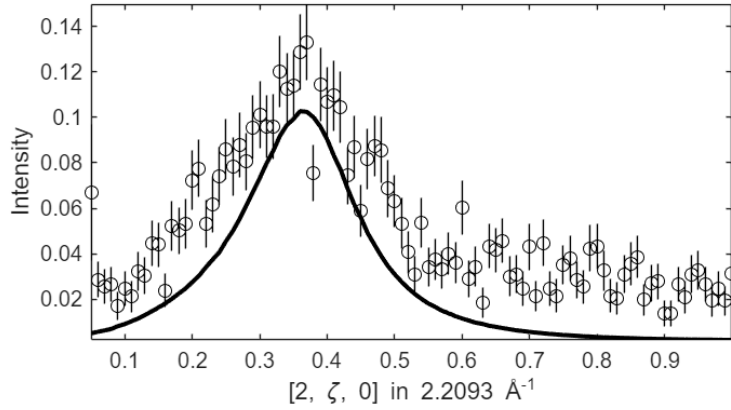


 **** En = 165±5 going to 215

Fe_ei401_align_sym3.sqw
 $S = 1.4 \pm 0.07$; $J_0 = 39 \pm 0.74$; $J_1 = 0 \pm 0$; $\gamma = 98.9128 \pm 3.88326$
 $0.1 \leq \xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, \eta]$, $160 \leq E \leq 160$
 $\zeta = -0.005:0.01:1.005$ in $[2, \zeta, 0]$

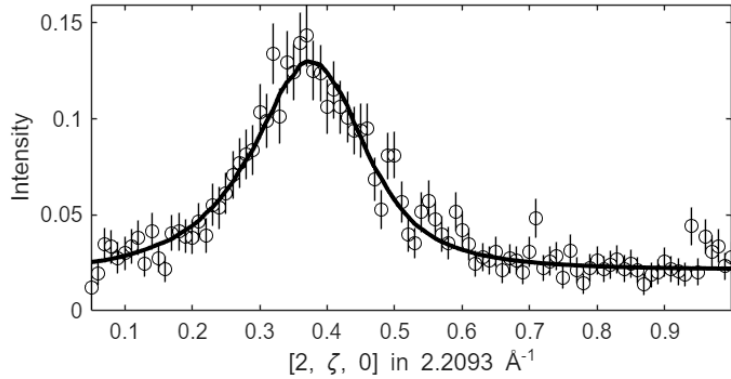


Fe_ei401_align_sym3.sqw
 $0.1 \leq \xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, \eta]$, $160 \leq E \leq 170$ eV
 $\zeta = -0.005:0.01:1.005$ in $[2, \zeta, 0]$

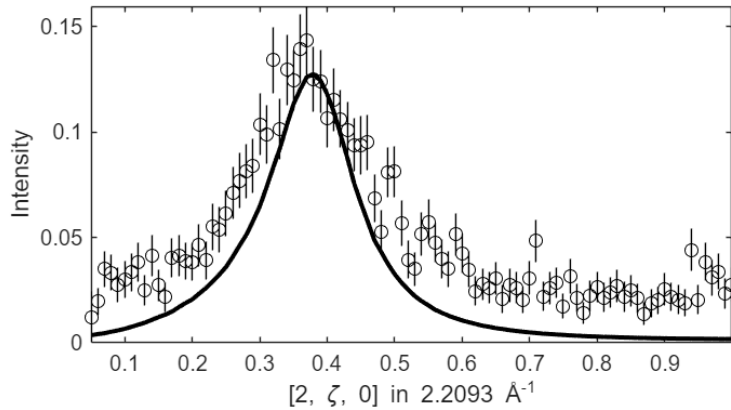


 **** En = 175±5 going to 215

Fe_ei401_align_sym3.sqw
 $S = 1.3 \pm 0.046$; $J_0 = 36 \pm 0.45$; $J_1 = 0 \pm 0$; $\gamma = 68.2762 \pm 2.33008$
 $0.1 \leq \xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, \eta]$, $170 \leq E \leq 185$ eV
 $\zeta = -0.005:0.01:1.005$ in $[2, \zeta, 0]$



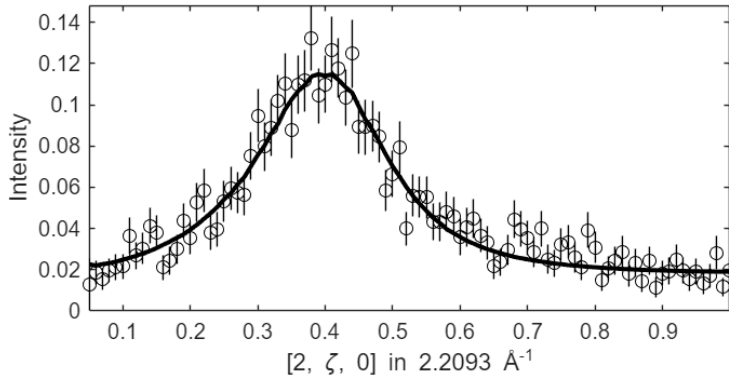
Fe_ei401_align_sym3.sqw
 $0.1 \leq \xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, \eta]$, $170 \leq E \leq 185$ eV
 $\zeta = -0.005:0.01:1.005$ in $[2, \zeta, 0]$



 **** En = 185±5 going to 215

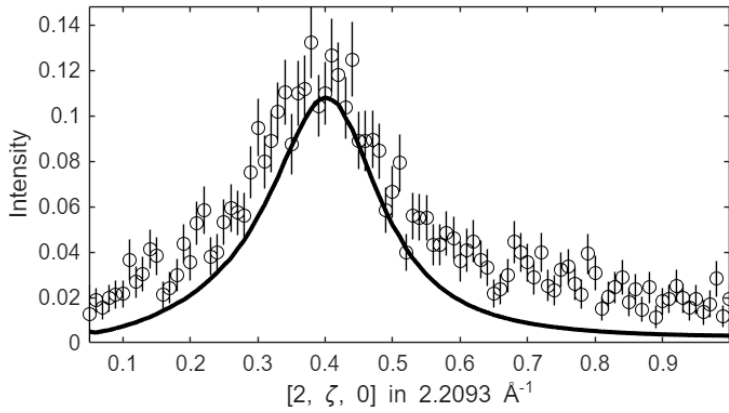
Fe_ei401_align_sym3.sqw

S= 1.5 ± 0.067 ; J0= 36 ± 0.47 ; J1= 0 ± 0 ; gamma= 94.1491 ± 3.87847
 $0.1 \leq \xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, \eta]$, $180 \leq E \leq$
 $\zeta = -0.005:0.01:1.005$ in $[2, \zeta, 0]$



Fe_ei401_align_sym3.sqw

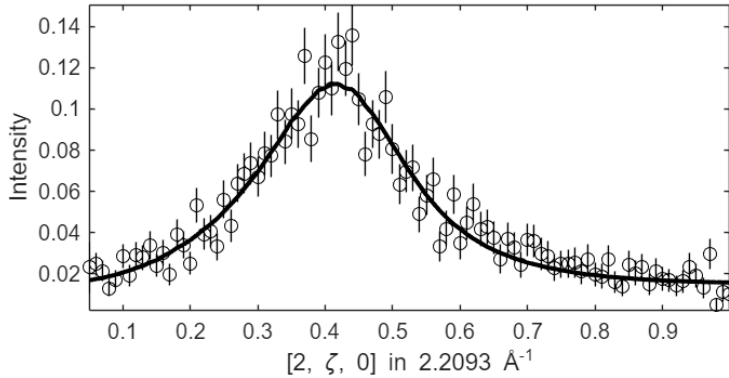
$0.1 \leq \xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, \eta]$, $180 \leq E \leq$
 $\zeta = -0.005:0.01:1.005$ in $[2, \zeta, 0]$



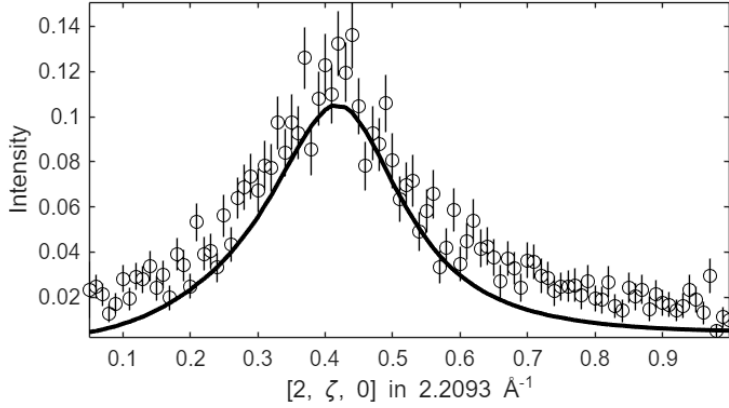
 **** En = 195 ± 5 going to 215

Fe_ei401_align_sym3.sqw

S= 1.9 ± 0.2 ; J0= 37 ± 1.3 ; J1= 0 ± 0 ; gamma= 127.379 ± 18.3668
 $0.1 \leq \xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, \eta]$, $190 \leq E \leq$
 $\zeta = -0.005:0.01:1.005$ in $[2, \zeta, 0]$

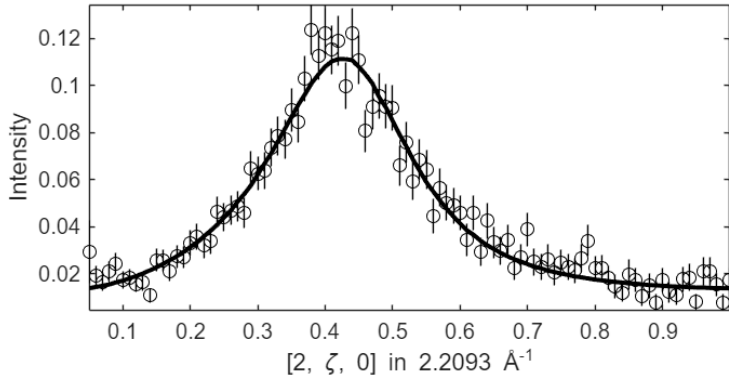


Fe_ei401_align_sym3.sqw
 $0.1 \leq \xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, \eta]$, $190 \leq E \leq$
 $\zeta=-0.005:0.01:1.005$ in $[2, \zeta, 0]$

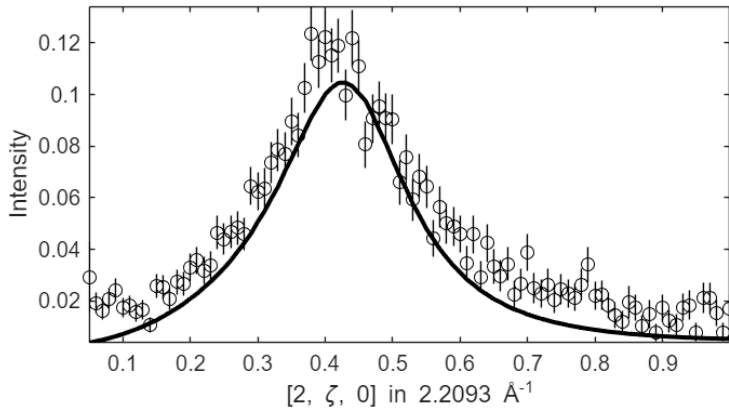


 **** En = 205±5 going to 215

Fe_ei401_align_sym3.sqw
 $S= 1.9\pm0.12$; $J_0= 37\pm0.61$; $J_1= 0\pm 0$; $\gamma=122.713\pm9.13092$
 $0.1 \leq \xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, \eta]$, $200 \leq E \leq$
 $\zeta=-0.005:0.01:1.005$ in $[2, \zeta, 0]$

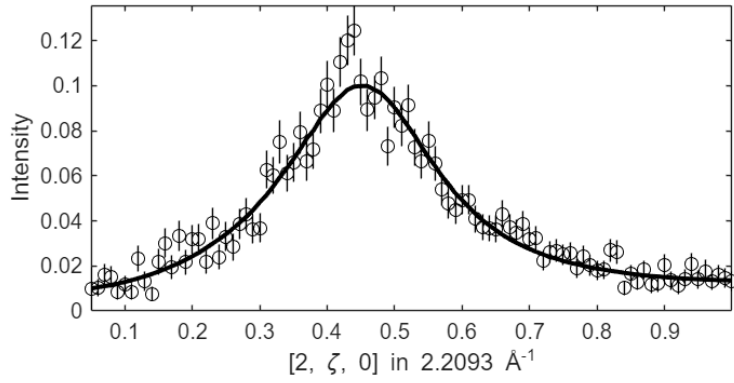


Fe_ei401_align_sym3.sqw
 $0.1 \leq \xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, \eta]$, $200 \leq E \leq$
 $\zeta=-0.005:0.01:1.005$ in $[2, \zeta, 0]$

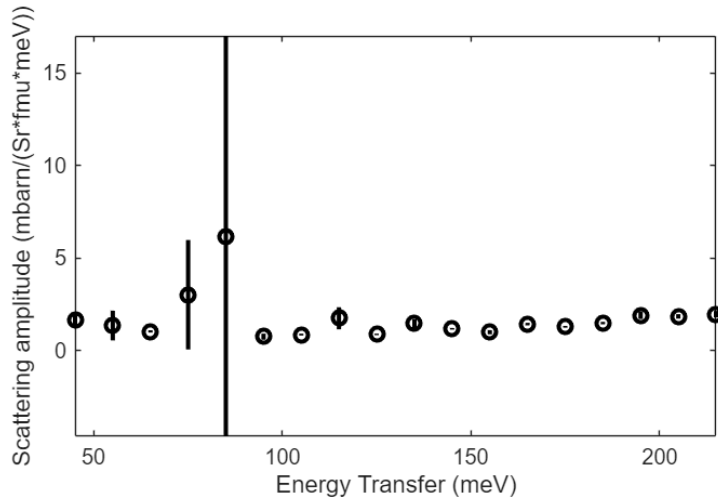
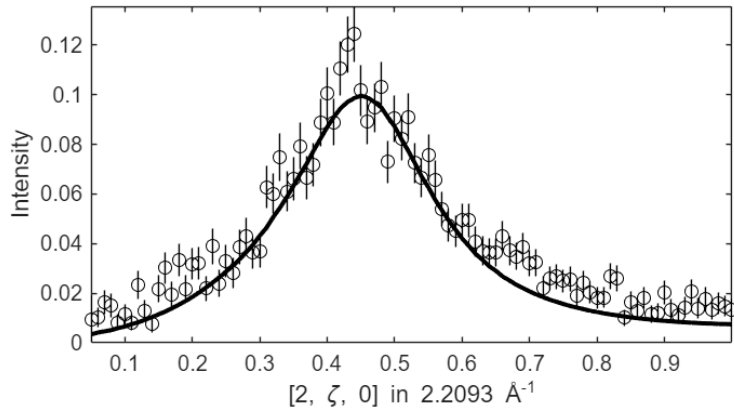


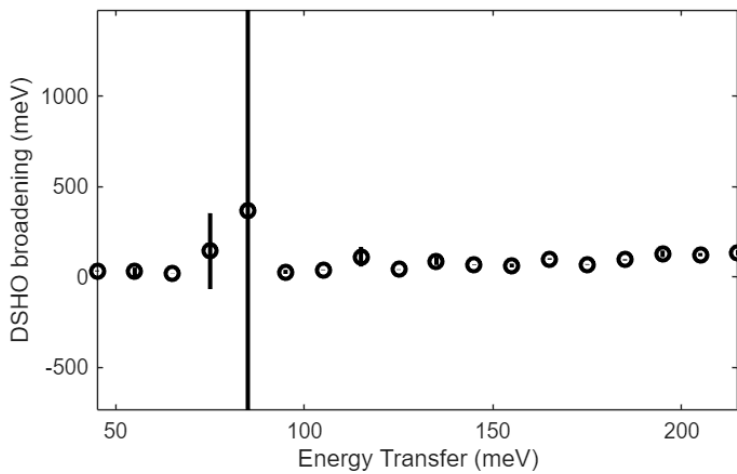
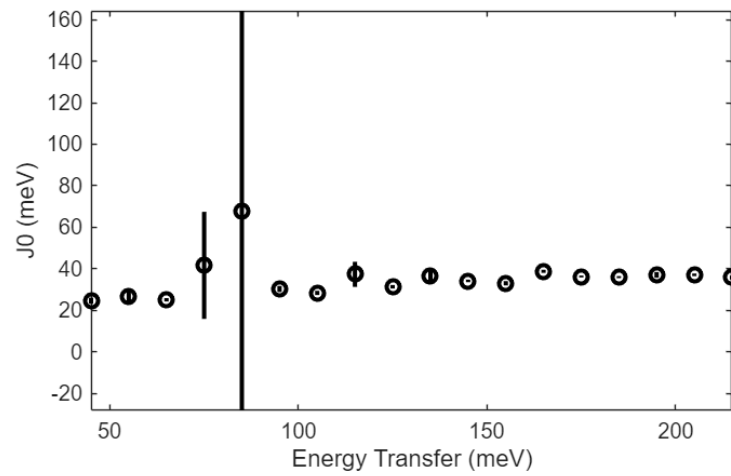
 **** En = 215±5 going to 215

Fe_ei401_align_sym3.sqw
 $S = 2 \pm 0.17$; $J_0 = 36 \pm 0.95$; $J_1 = 0 \pm 0$; $\gamma = 134.02 \pm 14.4519$
 $0.1 \leq \xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, \eta]$, $210 \leq E \leq$;
 $\zeta = -0.005:0.01:1.005$ in $[2, \zeta, 0]$



Fe_ei401_align_sym3.sqw
 $0.1 \leq \xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, \eta]$, $210 \leq E \leq$;
 $\zeta = -0.005:0.01:1.005$ in $[2, \zeta, 0]$





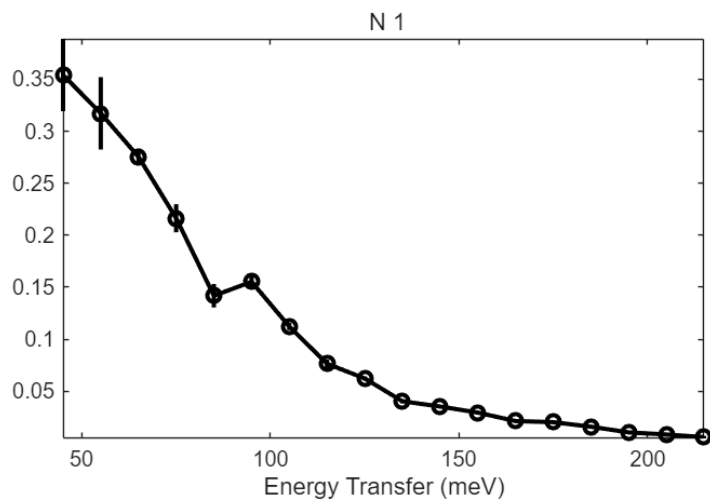
```

c1 = GH_FFCutsFits.(selectCuts);
GH_FFCutsFits.(selectCuts) = [c1{:}];

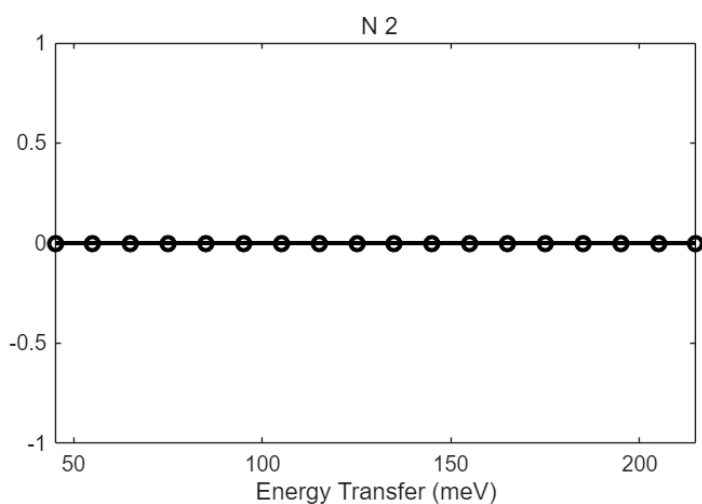
beep
beep
beep

all_fit_par = GH_FFCutsFits.([selectCuts, '_all_fit_par']);
bg_data = extract_bg_par(all_fit_par,[1,2]);
% original slope parameters
pd(bg_data(1));lx(all_fit_par(1).en,all_fit_par(end).en);keep_figure;

```



```
pd(bg_data(2));lx(all_fit_par(1).en,all_fit_par(end).en);keep_figure
```



```
%pd(bg_data(3));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_figure;
%pd(bg_data(4));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_figure
```

```
doc = matlab.desktop.editor.getActive;
doc.save;
thisFile = 'e:\SHARE\Fe\Data\analysis\06_fit_with_AQ0\fit_cutFe_GH.m';
targ_file = fullfile(fileparts(thisFile),['fit_cutFe_GH_',selectCuts,'.pdf']);
export(thisFile,targ_file);
```

```
fit_src_struct = GH_FFCutsFits;
save(CutFitFileGH,"fit_src_struct",-v7.3');
close all;
```

%end